

Barrel Timing Layer Assembly in CMS for the High Luminosity Update of the Large Hadron Collider

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Abstract:

The High Luminosity update for the Large Hadron Collider (LHC) at CERN (European Center for Nuclear Research) will allow for the detectors in the Large Hadron Collider to handle more data collection from particle collisions than previously, allowing for higher chances of detecting rare subatomic phenomena. For part of this update, the Barrel Timing Layer (BTL) was added within the Compact Muon Solenoid (CMS), which is one of the four detectors in the LHC. The BTL increases particle resolution, which makes studying individual particle collisions more efficient and much easier. Assembly of the Barrel Timing Layer first required assembling and testing for proper light yield/thermal cooling of sensor modules, detector modules, and cold plate trays. Afterward, 72 detector modules are installed onto each cold plate tray with 6 readout units that relay detector module data to computers for physics analysis. These trays are then tested for proper cooling and installed into a mockup of the BTL cylindrical aluminum frame. The BTL is made of 72 completed cold plate trays that line the inside of the cylindrical frame. The assembly process and its adjustment are smooth, but the testing of completed cold plate trays has manifested thermal issues (non-uniform tray cooling) in specific detector modules caused by thermal conductive material used in its copper housing. Further work includes changing this material and modifying the assembly process to make it more concise, efficient, and consistent.

Background:

The Large Hadron Collider at CERN is used to simulate the formation of the universe a few picoseconds after the Big Bang by colliding high-energy beams that create large quantities of particle collisions between hadrons (protons). Various particle showers result from each collision, which are recorded and analyzed by physicists to better understand subatomic phenomena and the standard model of particle physics. The Large Hadron Collider must be updated to handle a much higher luminosity (amount of particle collisions at once) to continue this journey of discovery. This update is called the High Luminosity Large Hadron Collider Phase (Phase 2 LHC), where each of the detectors in the particle collider is updated to handle collecting data from more particle collisions than before¹ (Benettoni, 2020). There are four collision points within the LHC. One of those collision points is at the Compact Muon Solenoid (CMS) which is a “general-purpose detector; that is designed to observe any new physics phenomena that the LHC might reveal.”² (CMS Detector, 2023). Within the CMS detector, “a new minimum-ionizing-particle precision timing detector (MTD) and a new luminosity detector will be installed”³ (Dabrowski et al., 2023).

The minimum ionizing particle timing detector (MTD) will detect the timeline of events more precisely regarding particle collisions. In more detail, “The precision time information from this MIP Timing Detector (MTD) will reduce the effects of the high levels of pileup expected at the HL-LHC, bringing new capabilities to the CMS detector”⁴ (Marzocchi, 2021). The MTD must be installed for phase 2 of the LHC to allow physicists to properly distinguish what particle interactions happened in each high-energy beam collision. Within the MTD is the Barrel Timing Layer (BTL) which detects the timing of individual particles entering/leaving the collisions used to determine the timing of particle collisions through the MTD. The Barrel Timing Layer is a thin, cylindrical detector that is made up of 72 trays that hold 144 detector modules⁴ (Marzocchi, 2021).

The construction/assembly of the BTL that CERN is responsible for includes assembling sensor modules, assembling detector modules, testing the light yield of sensor modules, testing the cooling properties of detector modules, installing the detector modules onto aluminum trays with CO2 cooling pipes, testing the cooling properties of the tray, and testing the installation of these trays into the BTST (the cylindrical chamber that contains the BTL) via a mockup BTST. CERN is also responsible for designing and constructing tools used to properly assemble sensor and detector modules as well as install these components onto 72 trays that line the BTST.

Results:

One fully assembled tray for the BTL has been made at CERN. This tray consists of 144 sensor modules, 72 detector modules, 6 readout units, and 288 Thermal Electronic Components (TECs) where each detector module has 4 TECs. The sensor modules, detector modules, and readout units for this tray were all assembled at CERN. When assessing the cooling of TECs in the final tray, the BTL lab group observed that some TECs were rapidly cooling compared to the

other TECs. All detector modules on the tray passed the TEC thermal tests individually, however. Thus, this was not in line with what we expected to happen, which was a tray with TECs that all work and cool down to 0 degrees Celsius. The lab went through an iterative testing process to isolate the issue with these TECs and investigate the reason for this rapid cooling. The concentrator cards and power concentrator cards that were attached to the detectors with rapid cooling TECs were replaced but did not cause a difference in testing. The software that generated the cooling plots for this test was not malfunctioning, as a thermal camera was used to confirm this rapid cooling effect. The detector modules with the malfunctioning TECs were disassembled, reassembled, retested individually for cooling properties, and then installed back onto the tray for another tray cooling test. This process only fixed some malfunctioning TECs, which means that the issue does not reside in the readout units or the cold plate itself but in the installation process of the tray.

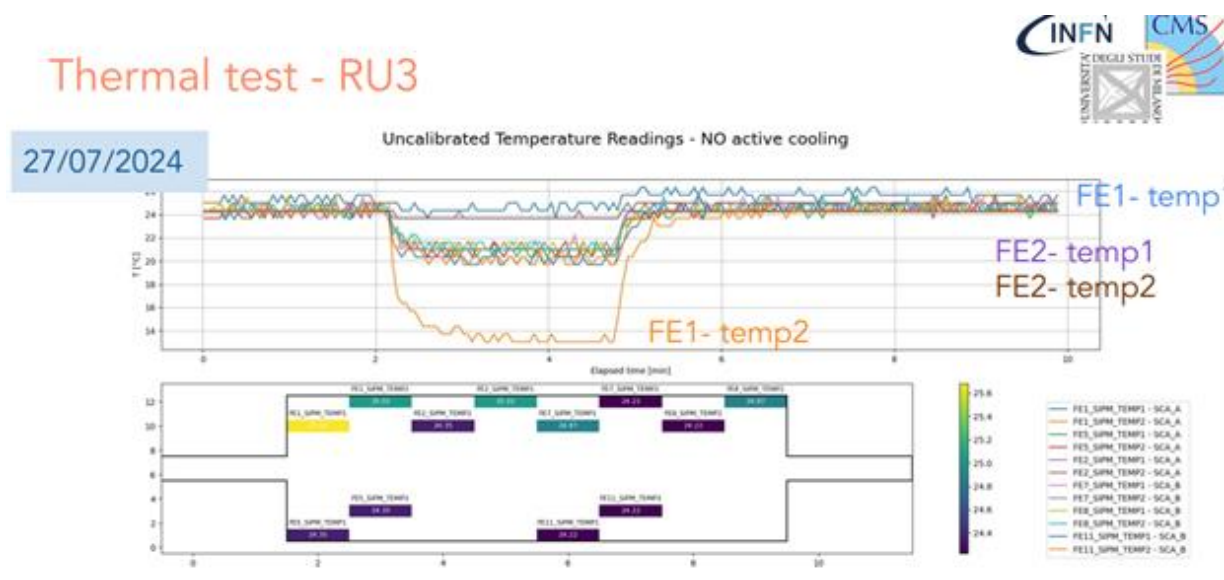


Figure 1: Plot of at least one of the tray readout units showing the temperature of TECs in Celsius over time in minutes and noticing the rapid cooling of one TEC. The term “FE” stands for Front End board and the number after it labels which FE the colored line is referring to on the plot (see key on bottom right). Each detector module has a front-end board to relay information about what it senses back to computers. The graphic on the bottom provides a map to the FE boards on the tray and a corresponding temperature gradient.

The lab deduced that this issue is caused by an electrical short in some of the detector modules, as the resistance measured between the malfunctioning TECs, and the cold plate was very low. When resistance is incredibly low, the current becomes remarkably high. This high current causes the affected TECs to become overpowered and cool down too quickly. The other TECs cannot receive as much current if most of it goes to a few TECs, which will cause other TECs to stay at the same temperature and not cool at all. If the current is too high for the electronic components, then an electrical short can happen, which in turn causes the above events

to manifest. Untightening the screws attaching the detector modules to the cold tray a little solved this issue partly. This indicated that faulty TEC electrical connections were being squished against electrically conductive material, and providing more space by loosening the screws could help solve this issue. The lab investigated what this electrically conductive material could be. The cause of the electrical short has been narrowed to the thermal material (T558) lining the copper housing around the detector modules. Some TECs are being squished against this material, which also happens to have an aluminum layer. This aluminum layer is electrically conductive, and if there is no space between the TECs and the thermal material it can cause a short.

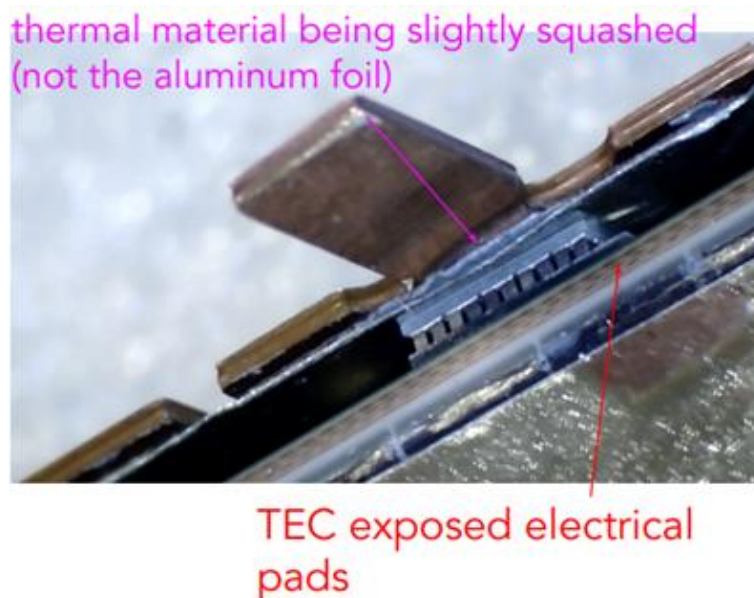


Figure 2: Photo of the cause of the shorts in some of the TECs.

The goal of this research over the summer is to refine/finalize the assembly and testing process for the BTL project. The need for refinement was emphasized through the malfunctioning of the first completely assembled BTL tray at CERN. Members of the BTL lab group running thermal tests on the completed tray needed to pinpoint the cause of rapid cooling and how to prevent it. The lab group worked on the disassembly, retesting, and reassembly of the shorting detector modules. Firstly, the lab group tested different thermal materials to line the copper housing. A detector module was taken off the assembled tray, and taken out of the copper housing, and a new thermal material was attached to the inside of the copper housing. From there, the detector module cooling test was run the same as before. This procedure was executed for each new type of thermal material. Out of T558, T557, T777, Kapton tape, and TOFHIR pads, the material with the highest thermal conductivity was T558. High thermal conductivity is crucial to facilitate detector module cooling. T558 is the same material used previously on the copper housings attached to the detector modules. Thus, the solution is not to replace the thermal material between the copper housing and the detector module.

BTL lab group members noticed that detector module cooling performance increased when there was less pressure between the copper housing and the detector module. This means that the detector module housing tool must be adjusted to apply less pressure between the housing and the module during its attachment. An updated design for the detector module housing tool was applied. The tightening screws that were used to pull apart the copper housing for inserting the detector module inside became cam levers. The cam levers were adjusted to pull apart the copper housing until there was a 2 mm radius bend at the bottom of the copper housing. The orientation of the L brackets on the DM housing tool was flipped to keep the claws parallel to the ground as they pulled apart the copper housing. It prevents unnecessary forces from acting on the claws.

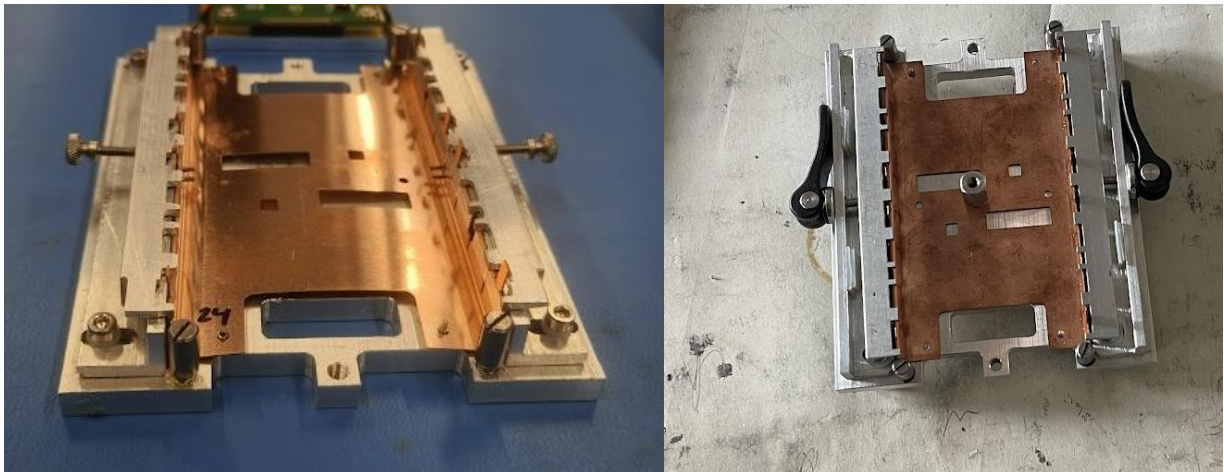


Figure 3: The left shows the original design for the DM housing tool. The right shows the updated DM housing tool with CAM levers that replaced the tightening screws and a backwards orientation of the brackets.

On a higher level of tray assembly, the cold box tests the thermal compliance of metal trays that will hold detector modules and readout units. The seal of the cold box needed redesign and repair. It was meant to guide the CO₂ pipes and dry air pipes to enter the cold box through one location and seal the inside of the box from the outside environment. The CO₂ pipes provide CO₂ to the cooling tubes on the tray so the tray can cool down, and the dry air pipes pump dry air into the cold box to prevent excessive humidity that condenses when the temperature is lowered to between -30 and -40 degrees Celsius. The seal reduces environmental variables when it comes to testing the cooling capabilities of the cold tray by sealing off the internal environment of the cold box. The previous seal broke due to its rigidity and wear on the material over time from the pipes that passed through it. The new design provided a stronger frame for the seal, less overall material use, more Airex for insulation, and plexiglass instead of plastic to hold the pipes in place as they enter the box. This design has not been tested, as the cold box test has not been executed since its construction and installation this summer. Already, the structural stability of the plexiglass is not strong enough to keep the CO₂ pipes in place. The top of the seal frame

made from metal framing rails keeps the CO₂ in place. Thus, the design will be improved so the plexiglass can keep the CO₂ pipes in place as they enter the cold box.

Methods:

Assembly of the BTL in the MTD of the CMS detector requires multiple stages of construction. The stages relevant to this report are assembling sensor modules, assembling detector modules, QA&QC testing of sensor and detector modules, assembling of cold trays, testing of cold trays, and installing detector modules onto these cold trays that will line the BTST to make the BTL. Members of the lab do not contribute to a particular task in this list, rather they contribute to each task at any given time. This allows for more efficiency, as the completion of a specific task is not reliant on one person. It also helps with refining the processes used to complete each task, and therefore creating higher quality construction for the HL LHC update in CMS.

Sensor Module Assembly:

The sensor modules are created by gluing two Scintillator Silicon Photomultipliers (SiPMs) to the sides of a LYSO crystal. This LYSO crystal is a combination of LYSO crystal rectangular prisms (or bars) whose faces correlate to the channels on the SiPMs. The gluing process must be uniform, precise, and without bubbles for the sensor modules to work properly. Bubbles in the glue as well as a non-uniform glue layer result in a lower light yield and/or significant inconsistency in light yield that the sensor module received. It is best to avoid gluing the SiPMs to the LYSO by hand and use a sensor module gluing tool designed by CERN. This allows for an even layer of glue to be applied to the SiPMs with a lack of bubbles and aligns the SiPM channels perfectly with the LYSO bars on the crystal.

Detector Module Assembly:

The detector modules are created by taking a detector module PCB board (already designed and constructed), placing two foam strips on the bottom of the PCB, attaching two sensor modules to the PCB board, and placing this into a copper housing. The copper housing is used to move heat away from the detector module itself allowing for rapid cooling. The two foam strips act as cushioning between the LYSO crystals on the sensor modules. The detector module housing tool was designed and constructed at CERN to pull apart the copper housing the detector module resides in for its installation. Currently, there are at least 10 detector module housing tools constructed and used for this purpose at CERN and at universities that work with the BTL.

Sensor Module Light Yield Tests:

Before the sensor modules are used in assembling detector modules, they are tested for their light yield content and consistency. When the LYSO is hit by particles, it absorbs them and emits light which reaches the SiPMs for detection. There is a tool called the QA/QC, which is contained in a black box. It is created from various PCB boards connected to an Arduino which was programmed to read the light yield of all the sensor modules installed on the QA/QC. The sensor modules would be installed onto these PCB boards for testing. The graphs presented after the QA/QC test runs show the light yield detected from the left SiPM, right SiPM, and the average between the two for each channel on the SiPMs. If the light yield is high, and the graphs show little to no variance between the left and right SiPM, then the sensor module is ready to be used in constructing a detector module. If not, then there was inconsistency with gluing the SiPMs to the LYSO crystal, the SiPM was broken, or the LYSO crystal was broken.

Detector Module Cooling Tests:

Detector modules are also tested for their thermal cooling properties through Arduino. The detector module is connected to thermal pads that are linked to an Arduino programmed to detect the temperature of the detector module itself as well as its copper housing over an extended period. The thermoelectric coolers on the detector modules are then activated during testing to cool the detector modules. The lab discovered that the cooling plateaus after 5 minutes, so that is the length of these tests. The goal of these tests is to see how cool each thermoelectric cooler can be after 5 minutes, and how hot the copper housing is due to pulling heat away from the detector module. The test also checks for consistency amongst the thermoelectric coolers. So far, the thermoelectric coolers have changed the temperature of the detector modules by about 25 degrees Celsius, and the copper housing reached temperatures of 40 to 45 degrees Celsius on average. The goal is to cool the detector modules as much as possible to prevent radiation damage from extensive use in the particle collider.

The first phase of assembling the BTL involved building sensor and detector modules and testing their light yield and cooling properties. The second phase required assembly at a larger scale. It includes assembling the cold plate trays, installing the detector modules onto these trays, and installing the necessary electrical components for data collection from the detector modules. These electrical components are 6 readout units, which consist of 6 concentrator cards, 6 power concentrator cards of 1.2 volts, and 6 power concentrator cards of 2.5 volts. Testing the cold plate tray at each stage of its assembly is critical to ensure proper function within the BTL in the HL LHC update.

Cold Tray Assembly:

First, the cold plate tray needs to be assembled and tested. The aluminum cold plates come in 6 sections, each labeled CP1, CP2, and CP3 where CP stands for “cold plate”. CP1 and CP2 are used for the ends of the cold plate and 4 CP3 plates are used for the middle. The reason each cold plate is labeled differently has to do with the placement of screw holes on the plates for detector module assembly and the slots that accommodate the cold tubes that cool the tray. Each

plate is then placed on a cold plate frame that holds the cold plate together during its assembly. Once that is finished, three alignment pins are placed in between each plate with a 0.2 mm aluminum spacer, and the plates are pushed together to make the base of the cold tray.

Once the cold plate is in one piece, the thermal paste is applied to the cold plate tube slots. MX-4 thermal compound is used for the thermal paste, as it has high thermal conductivity to dissipate heat and is non-electrically conductive⁵ (Artic, 2023). Thermal paste is applied using a thermal paste applicator jig. The jig aligns 4 thermal paste tubes to the 4 cold plate tube slots and squeezes out thermal paste from each tube at a controlled rate as it moves across the cold tray. set to a velocity of 800 mm/min. Four MX-4 thermal compound tubes are placed into the orange tube holder and the plungers are adjusted to allow a bit of thermal paste to start leaving each tube. The applicator jig is then tilted downwards so the syringes touch the 4 corresponding groves in the cold plate. The start is pressed on the control box after setting the velocity, and the applicator jig applies the thermal paste to the cold tube groves as it moves down two conveyor belts on each side.

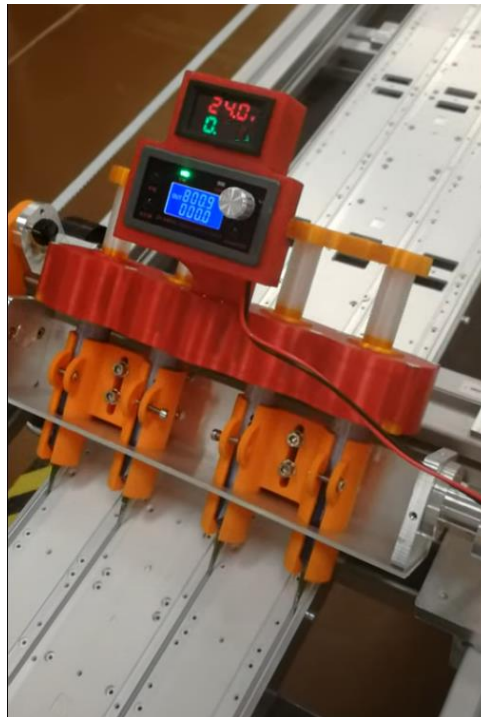


Figure 4: Picture of thermal application jig with 4 MX-4 thermal tubes installed, moving down conveyor belts above the cold plate supporting frame to apply thermal paste into the cold tube groves.

A 3D-printed smoothing handle is placed in one of the cold tube groves and pushed from one side of the cold plate to the other to evenly coat the thermal paste in the cold tube groves while removing excessive thermal paste. Every cold tube groove goes through this process. Since

this part is done by hand, there is an inconsistency in the amount of thermal paste in each section of the cold tube grooves. Therefore, this process may be automated in the future like the thermal paste application process.

A cold plate tube set is carefully lifted and placed into these grooves. To secure the cold tubes to the plate, aluminum top pieces called lamina are placed over the cold tubes and screwed into the cold plate. Once this is finished, the cold plate is attached to supporting framing rails for storage until installation into the BTL. This is done using aluminum attachments that screw into the cold plate and into the tops of these framing rails. The configuration is then lifted off the cold plate frame where it was assembled and is placed to the side, indicating that the assembly is now finished.



Figure 5: Two assembled cold plates are attached to their supporting frames.

Cold Tray Testing:

Once the cold plate is assembled, it needs to be tested for thermal compliance in the BTL. Two tests are conducted on each cold plate. The first test requires placing the cold plate onto a heating pad of its size. Black insulating tape is placed to completely cover the top side of the cold plate. All lights near the area are turned off, and then a thermal camera is used to look at the thermal consistency of the cold plate. The goal is to make sure that the cold plate consistently stays at 20 degrees Celsius and no higher. If some parts of the plate are warmer than others, then that highlights a potential issue with the cold plate, cold tubes, lack of black tape coverage, or issue with the validity of this test. The lab has worked on debugging the testing procedure for detecting a consistent thermal gradient in the cold plate.

The second test collects data on how well the cold tubes cool down the cold plate. The cold tubes should ideally cool the plate to -30 degrees Celsius inside a cold box. The cold box is created from thermally insulating material where the CO₂ supply for the cold tubes and cold dry

air tubes enter inside. These pipes are sealed off as well as possible to prevent the outside environment of the lab from affecting the cool internal temperature of the cold box. Once the cold plate is connected to the CO₂ supply and the cold dry air supply, the cold box is placed over it and anchored onto the table. When the dry air supply is turned on, and the box has cooled for a couple of days, the CO₂ supply is turned on and data is collected on the cold plate temperature. If the cold plate reaches a temperature of -30 degrees Celsius from 0 degrees Celsius inside the cold box, then the cold plate is working properly.



Figure 6: The cold box is secured over a cold plate tray for testing. The black tubes are CO₂ supply, and the thin capillary tubes are for dry air to enter and cool the tray.

Once the cold plate has passed both tests, it is ready for detector module installation. The cold plate tray is placed and locked into the assembly test stand. The cold plate tray can now be rotated to any 45-degree increment to make detector module installation easier. Firstly, the readout units must be installed onto the tray to relay the data taken by detector modules to a computer. These readouts consist of concentrator cards and power concentrator cards installed on the top of the tray. Thermal pads are placed over the electronic components of each readout unit facing down onto the tray. Afterward, each of the 6 readout units is screwed onto the 6 cold plate sections of the tray, respectively. Each detector module has board ID switches on its front-end board which are adjusted to correspond to its location on the tray and which readout unit it will attach to. Once the detector module board ID switches are prepared, two people will help install them onto the tray. One person will hold the detector module against the bottom side of the board in its corresponding section of the tray, and the other person will use 6 screws to attach it to the tray on the front side. This is what the final tray should look like once computer/electronic cables are plugged into the tray.

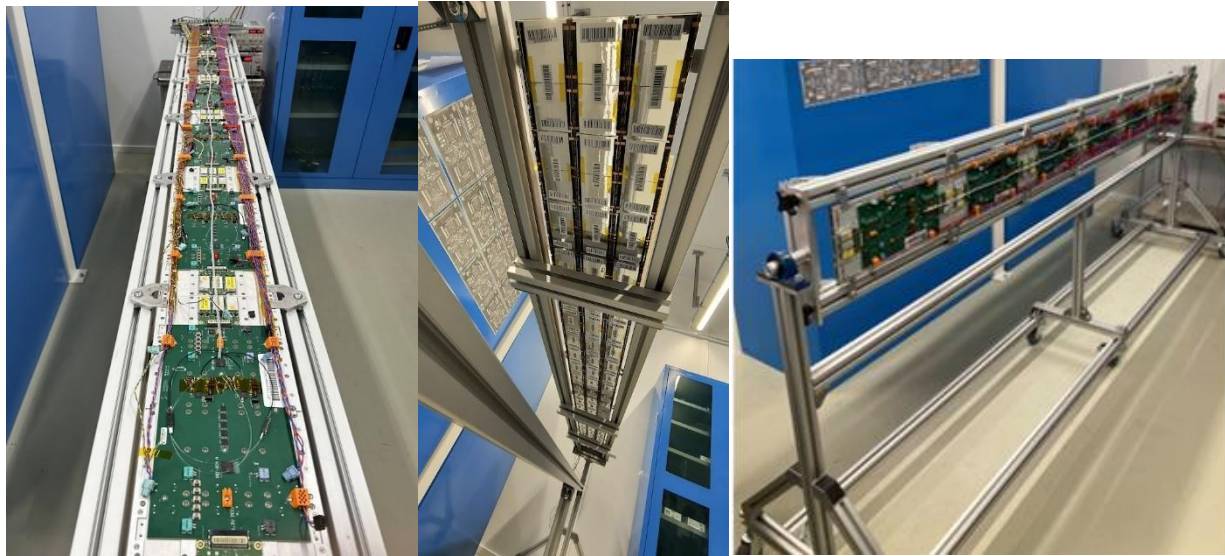


Figure 7 (left): The cold plate tray frame (top part tilted 90 degrees) is installed into the assembly test stand and locked into the assembly test stand using black knobs.

Figure 8 (right): The two pictures of the fully assembled tray top and bottom.

The final tray assembly must be tested for thermal compliance, even though each component was tested before. This ensures there are no issues with the assembly process, and all components work together smoothly. Firstly, the TECs in the detector modules are activated to cool down the tray. The TECs' goal is to uniformly cool down the tray from room temperature to 0 degrees Celsius. Once the tray assembly passes this test, its cold tubes are tested for cooling properties alongside the TECs. The tray assembly is moved to the cold box location for testing. The TECs and the CO₂ supply are activated, and the cold box is sealed over them. This time, the cold box inner temperature is not cooled down to 0 degrees Celsius, as this is the job of the TECs on each detector module. The tray is expected to uniformly cool from room temperature to -30 degrees Celsius (Benettoni, 2020).

Refining the BTL Tray Assembly Process: Cold Box Seal

The old cold box seal broke over the summer and needed redesigning and replacement. The old cold box seal could not withstand the forces of the CO₂ pipes that it held in place. The seal must be a structurally sound threshold for the CO₂ and dry air pipes while concealing the inside testing environment from outside factors such as dust, warm temperatures, and humidity.

The old cold box seal was a rectangular prism that had dimensions 115 mm in length, 70 mm in width, and 145mm in height. The rectangular prism was made from multiple 3D printed layers with dimensions of 91 mm in length, 70 mm in width, and height varying between 10.6 mm to 41.5 mm. Some of these layers had semicircular holes in them that accommodated for the CO₂ pipes and dry air pipes that would enter the cold box. The bottom most layer had clearance holes for M4 by 30 mm screws that secured the seal into the table where the cold box setup resides.

Lastly, there were four dove joints sticking out from each layer (two on the left and two on the right) that connected 3D printed sides of the seal together. When the three sides (top, left, and right) are screwed together, they combined each of the layers together to make the seal. The layers can be rearranged within the seal to adjust where the pipe cross through, which can accommodate varying setups of the cold tray in the cold box.

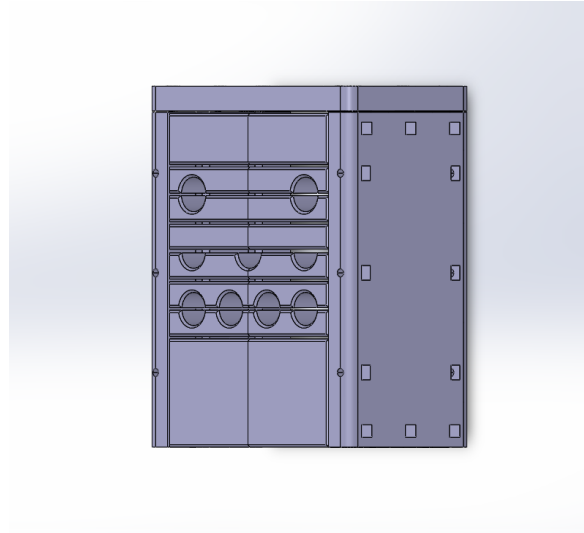


Figure 9: The Computer Aided Design Model of the old cold box seal design in SolidWorks

The new seal design replaces the sides of the seal with 20 x 40 mm t-slot framing rails. Aluminum is more structurally stable than PLA or any other plastic typically used for 3D printing. The bottom most framing rail has clearance holes for the M4 x 30 screws that attach the seal to the table. The framing rail sides of the seal, known as the outer frame of the seal, are attached using 8 t-slot corner brackets. The seal layers were laser cut from plexiglass instead of 3D printed. These layers had length of 115 mm, width of 2 mm, and heights that vary from 11mm to 16.5mm. Each layer had two clearance holes for M4 x 8 screws to attach the layers to the framing rails that made the frame of the new seal. In between the layers were slices of Armor Flex Foam to insulate the pipes and the entrance/exit to the cold box. The figure below demonstrates the design.

The new frame of the seal was more structurally stable than before, as it did not bend to accommodate for the pipes that crossed through. The plexiglass was durable enough to hold the dry air pipes in place as they crossed through the seal. However, the plexiglass could not hold the CO₂ pipes in place. One of the plexiglass layers meant to hold the CO₂ pipes fractured completely in half during installation. This was the only plexiglass layer meant to hold the CO₂ pipes in place that was installed. It was clear that 2mm thin plexiglass did not provide enough resistance to guide the CO₂ pipe through the small area where the seal was into the box. Thus, the CO₂ pipes crossed through the seal at the very top so the top framing rail could hold the CO₂ pipes in place. This temporary solution appears to work, as the seal is structurally stable, and the

CO2 pipes are held in place as they cross into the cold box. The cold box test has not been run with the new cold box seal yet, so the integrity of the design is unknown. In the future, either thicker plexiglass will be used for the layers, or the cold box seal will be redesigned.

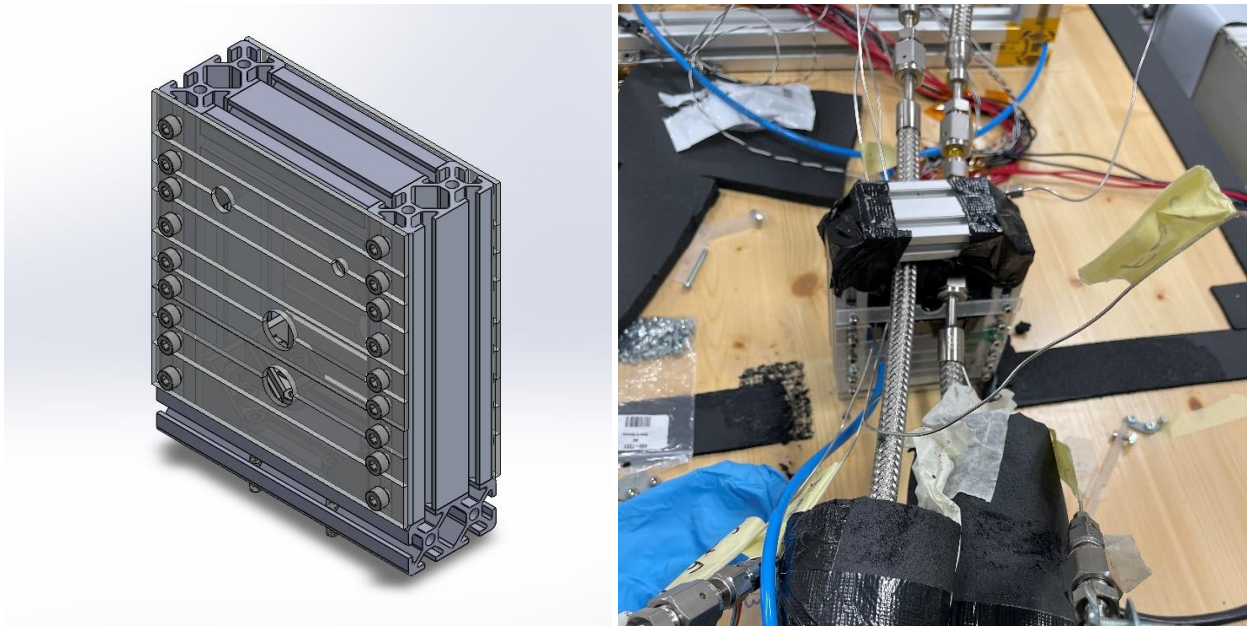


Figure 10: The left shows the new cold box seal design in SolidWorks, and the right shows the installation of the new cold box seal.

Acknowledgements:

I would like to share my utmost appreciation for my colleague Guillermo Reales Guterrez for guiding my work in the BTL assembly process and for his availability to discuss and collaborate with any task I was working on. He scheduled my training for the use of the Epilog laser cutting machine and milling machines at CERN. We worked together on designing the new cold box seal and on the general tray assembly. If I needed help with my work, he was a reliable source.

I am truly thankful to Adi Bornheim and Maria Spiropulu for providing the opportunity to work in person at CERN on the BTL assembly project for the High Luminosity Large Hadron Collider Update. This opportunity was a rich learning experience, from understanding the assembly procedure for all aspects of the BTL to witnessing firsthand the international collaboration conducted for BTL that extends beyond borders. I also had the opportunity to learn about each aspect of the CMS detector, and how the BTL would be integrated with the rest of the detector in the HL update.

I would like to voice my gratitude to Brian Cardwell, as he was tremendous in providing my orientation and welcoming me into the BTL group at CERN. He taught me how to assemble sensor modules and detector modules. He also consistently utilized stellar communication skills while at CERN and the University of Virginia later in the summer. I learned the importance of documentation, even for the smallest of steps, for the assembly of powerful yet delicate tools at CERN such as the BTL. He also displayed such kindness and community towards everyone in the BTL lab group and provided wonderful memories of working and having dinner with the group.

I immensely appreciate Ianos Schmidt's and Arjun Herring's consultation on the DM housing tool design and manufacturing process. While Ianos Schmidt has many tasks to complete as a full-time mechanical engineer in the BTL project, he was available to adjust the design of the DM housing tool based on feedback from the BTL lab group. I collaborated with him on changes to the design, and the plan to manufacture one updated DM housing tool for each BTL assembly location around the world. Arjun Herring helped me gather the tools I needed to work on making DM housing tools and showed me how to use these tools. He was also supportive of my progress on each project I progressed over the summer.

I would like to extend my gratitude to Alessio Ghezzi and Simona Palluotto for their well-documented and clear feedback on the previous DM housing tool, and their ideas for improvements to implement in the newest design. The effort Alessio Ghezzi and Simona Palluotto gave by iteratively testing each aspect of the currently assembled BTL tray to discover its reason for malfunctioning was critical work. It was critical to find that the DM housing tool was creating detector modules with their housings attached too tightly. We learned that the tool needs adjusting to prevent this issue from arising in the future and to properly reconstruct the malfunctioning detector modules. I also admire their supportive and uplifting attitude during the

failure of the first completely assembled BTL tray. This motivated me to persevere when these events occur, and that failure is part of a process, not the end of a project.

Lastly, I would like to thank my coworkers Noor Abduljalil J Abdulla, Brandon Frank Peterson, and Alex Mayanka Albert for their kind and collaborative attitudes during the BTL assembly process. Each one of them often asked if I needed assistance with any of my tasks during the summer. I also could ask them questions about aspects of the BTL assembly they worked on. They each explained their work in-depth and were patient with my learning process. I was able to better understand the BTL assembly and testing process, as well as its integration into CMS. They were also wonderful company for eating lunch.

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